

1. What are off-the-shelf (OTS) software products?

- A) Software products that require extensive development activities.
- B) Software products available for any user, at cost or not, and used without the need to conduct development activities.
- C) Software products developed and tested independently of Software Solutions.
- D) Software products that provide reusable functionality.

2. Which ISO standard provides a hierarchical framework for quality definition?

- A) ISO-27001
- B) ISO-9126
- C) ISO-9001
- D) ISO-14001

3. What is the top-level quality characteristic of functionality?

- A) The ease of use from a user viewpoint.
- B) The capability of software to maintain its level of performance under stated conditions.
- C) The essential purpose of any product or service.
- D) The relationship between the level of software performance and the amount of resources used.

4. What is the top-level quality characteristic of reliability?

- A) The essential purpose of any product or service.
- B) The relationship between the level of software performance and the amount of resources used.
- C) The capability of software to maintain its level of performance under stated conditions for a stated period of time.
- D) The ease of use from a user viewpoint.

5. What is the top-level quality characteristic of usability?

- A) The relationship between the level of software performance and the amount of resources used.
- B) The ease to use any function especially from user viewpoint.
- C) The essential purpose of any product or service.
- D) The capability of software to maintain its level of performance under stated conditions.

6. What is the top-level quality characteristic of efficiency?

- A) The relationship between the level of software performance and the amount of resources used, under stated conditions.
- B) The essential purpose of any product or service.
- C) The capability of software to maintain its level of performance under stated conditions.
- D) The ease of use from a user viewpoint.

7. What is the top-level quality characteristic of maintainability?

- A) The capability of software to maintain its level of performance under stated conditions.
- B) The ease of use from a user viewpoint.
- C) The effort needed to make specified modifications.
- D) The essential purpose of any product or service.

8. What is the top-level quality characteristic of portability?

- A) The ability of software to be transferred from one environment to another.
- B) The essential purpose of any product or service.
- C) The capability of software to maintain its level of performance under stated conditions.
- D) The ease of use from a user viewpoint.

9. What is an error?

- A) An incorrect condition in a computer program.
- B) A semantic mistake made by a programmer.
- C) A syntactic mistake made by a programmer.
- D) A human action that produces an incorrect result and/or the mistakes made by a programmer.

10. What is a defect?

- A) An incorrect condition in a computer program.
- B) The result of a human action that produces an incorrect result.
- C) The deviation from customer requirements.
- D) The mistakes made by a programmer.

11. What is a bug?

- A) An error found after the software is shipped into production.
- B) An error found during the testing phase.
- C) An error found before the software is shipped into production.
- D) An error found during the coding phase.

12. What is a fault?

- A) An incorrect step, process, or data definition in a computer program.
- B) An error found during the coding phase.
- C) An error found before the software is shipped into production.
- D) An error found after the software is shipped into production.

13. What is a failure?

- A) The presence of defects in a software solution.
- B) A result of a human action that produces an incorrect result.
- C) The inability of the program to behave as expected within given performance requirements.
- D) An incorrect step, process, or data definition in a computer program.

14. What is the result of a failure?

- A) The delivered service no longer complies with the specifications.
- B) The presence of defects in a software solution.
- C) The inability of the program to behave as expected within given performance requirements.
- D) The deviation from customer requirements.

15. What example is given for an error?

- A) Examine the following lines of code:
- B) Let's assume a software solution for online payments.
- C) An incorrect step, process, or data definition in a computer program.
- D) Due to wrong sign there is deviation from expected result.

16. What example is given for a defect?

- A) Let's assume a software solution for online payments.
- B) The set of attributes related to the capability of software to maintain its level of performance under stated conditions for a stated period of time.
- C) An incorrect step, process, or data definition in a computer program.
- D) Software that can be transferred from one environment to another.

17. What example is given for a bug?

- A) An incorrect condition in a computer program.
- B) July 28, 1962 -- Mariner I space probe.
- C) Bugs are the errors found before the software is shipped into production.
- D) An error found during the testing phase.

18. What example is given for a fault?

- A) Let's assume that the requirement is to write a program to add two numbers.
- B) An incorrect step, process, or data definition in a computer program.
- C) An error found during the coding phase.
- D) An error found before the software is shipped into production.

19. What example is given for a failure?

- A) An error found before the software is shipped into production.
- B) The deviation from customer requirements.
- C) An incorrect step, process, or data definition in a computer program.
- D) Due to wrong sign there is deviation from expected result.

20. What is the purpose of functionality?

- A) To have a relationship between performance and resources used.
- B) To provide and perform desired functions.
- C) To maintain its level of performance under stated conditions for a stated period of time.
- D) To be easily used from a user viewpoint.

21. What is the purpose of reliability?

- A) To maintain its level of performance under stated conditions for a stated period of time.
- B) To provide and perform desired functions.
- C) To be easily used from a user viewpoint.
- D) To have a relationship between performance and resources used.

22. What is the purpose of usability?

- A) To maintain its level of performance under stated conditions for a stated period of time.
- B) To be easily used from a user viewpoint.
- C) To have a relationship between performance and resources used.
- D) To provide and perform desired functions.

23. What is the purpose of efficiency?

- A) To have a relationship between performance and resources used.
- B) To maintain its level of performance under stated conditions for a stated period of time.
- C) To provide and perform desired functions.
- D) To be easily used from a user viewpoint.

24. What is the purpose of maintainability?

- A) To provide and perform desired functions.
- B) To maintain its level of performance under stated conditions for a stated period of time.
- C) To make specified modifications.
- D) To be easily used from a user viewpoint.

25. What is the purpose of portability?

- A) To be transferred from one environment to another.
- B) To maintain its level of performance under stated conditions for a stated period of time.
- C) To be easily used from a user viewpoint.
- D) To provide and perform desired functions.

Answer Sheet

1. B
2. B
3. C
4. C
5. B
6. A
7. C
8. A
9. D
10. C
11. C
12. A
13. C
14. A
15. A
16. A
17. B
18. A
19. D
20. B
21. A
22. B
23. A
24. C
25. A

1. What is the strategy to identify root causes of defect and prevent them from recurring?

- A) Defect Detection
- B) Defect Prevention
- C) Defect Containment
- D) Defect Removal

2. Which activity helps to improve the quality of the software product?

- A) Defect Removal
- B) Defect Detection
- C) Defect Containment
- D) Defect Prevention

3. What is the process to minimize defects during software development?

- A) Defect Detection and Reduction
- B) Defect Removal
- C) Defect Containment
- D) Defect Prevention

4. What is the unrealistic expectation in software development?

- A) Project or product with 0 bug count
- B) Project with increased bug count
- C) Random occurrence of bugs
- D) Project with reduced bug count

5. Which technique aims to remove as many injected faults as possible?

- A) Defect Removal or Containment
- B) Defect Prevention
- C) Defect Detection
- D) Defect Containment

6. Why is defect containment necessary in software development?

- A) To prevent failures caused by rare condition defects
- B) To increase the number of detected defects
- C) To reduce the effort in fixing defects
- D) To introduce controlled defects

7. Which module discusses the cost of software quality?

- A) Module 04: Software Quality Assurance
- B) Module 03: Software Quality Metrics
- C) Module 02: Cost of Software Quality
- D) Module 01: Introduction to Software Quality

8. What makes defining software quality difficult?

- A) Lack of available software quality frameworks
- B) Limited understanding of software development
- C) Inconsistent definition of software features
- D) Different perceptions and definitions among stakeholders

9. What is the historical failure rate of software?

- A) Highest failure rate among all products
- B) Low failure rate compared to other products
- C) Equal to the failure rates of other products
- D) No historical failure rate recorded

10. What are the economic consequences of low software quality?

- A) Serious economic losses
- B) Increased profitability
- C) Decreased competition
- D) Improved market reputation

11. What is the key factor impacting user satisfaction in software products?

- A) Overall behavior of the system
- B) Software development process
- C) Efficiency of the software team
- D) Number of features

12. What is the high concern in software quality engineering?

- A) Budget allocation
- B) Technical feasibility
- C) Project timeline
- D) User satisfaction

13. How is quality of a software product defined?

- A) Through features and quality
- B) Through budget and timeline
- C) Through management decisions
- D) Through user opinions

14. What determines the features and quality of a software product?

- A) Development team size
- B) Requirements
- C) Technology stack
- D) Marketing strategy

15. What are the tangible economic values associated with software quality?

- A) Average software quality
- B) Exceptional software quality
- C) Better than-average software quality
- D) Poor software quality

16. What are the natural enemies in development projects?

- A) Productivity and efficiency
- B) Functionality and quality
- C) Technology and security
- D) Budget and timeline

17. In a fixed budget project, what happens when the number of features is increased?

- A) Increased budget for quality maintenance activities
- B) Budget reduction for both features and quality
- C) Less budget for quality maintenance activities
- D) No impact on quality activities budget

18. What is the level of investment in the Function-Quality-Cost model?

- A) F
- B) A & B
- C) None of the above
- D) Q

19. Which factor among Quality, Function and Cost is determined by the requirements?

- A) Quality
- B) Function
- C) Cost
- D) All of the above

20. What does the Cost Breakup denote in the Quality vs. Pre-defined Budget scenario?

- A) Testing Cost and Documentation Cost
- B) Development Cost and Quality Cost
- C) Support Cost and Marketing Cost
- D) Hosting Cost and Maintenance Cost

21. Why are companies under pressure to deliver more and ignore quality assurance?

- A) To maximize their profits without considering quality
- B) To avoid delays in project completion
- C) To meet the demands while staying within the fixed budget
- D) To fulfill legal requirements

22. In a fixed-priced budget project, what happens to quality when the number of features is increased?

- A) Quality decreases
- B) Quality increases inversely
- C) Quality remains the same
- D) Quality increases proportionally

23. What is the term used to describe the decrease in quality due to increasing features in a closed-budget project?

- A) Capability-Efficiency-Value
- B) Performance-Security-Budget
- C) Feature-Quality-Price
- D) FunctionQuality-Cost

24. What is the total budget in the Quality vs. Pre-defined Budget scenario?

- A) PKR 80,000
- B) PKR 40,000
- C) PKR 100,000
- D) PKR 60,000

25. What is the cost per feature in the Quality vs. Pre-defined Budget scenario?

- A) PKR 25,000
- B) PKR 15,000
- C) PKR 5,000
- D) PKR 10,000

26. What is the purpose of quality maintenance activities in a closed-budget project?

- A) To ensure the desired level of quality for the product
- B) To reduce the overall development cost
- C) To enhance the software features
- D) To speed up the project delivery

27. What is the long term damage caused by ignoring quality control?

- A) Inflated development costs
- B) Unstable software with frequent failures
- C) Decreased user satisfaction
- D) Delayed project delivery



Answer Sheet

1. B
2. D
3. A
4. A
5. A
6. A
7. C
8. D
9. A
10. A
11. A
12. D
13. A
14. B
15. C
16. B
17. C
18. B
19. A
20. B
21. C
22. A
23. D
24. C
25. A
26. A

1. What is the impact of missing quality requirements in software products?

- A) Higher market share
- B) Increased budget
- C) Massive losses
- D) Improved user interface

2. Which of the following is an example of direct cost of missing quality?

- A) Financial loss
- B) Investment in research and development
- C) Damage to reputation
- D) Loss of market share

3. What is the main difference between direct and indirect costs of missing quality?

- A) Direct costs occur immediately after an event, while indirect costs occur after a long time
- B) Direct costs are financial losses, while indirect costs are non-financial losses
- C) Direct costs are easily observable/measurable, while indirect costs are difficult to calculate
- D) Direct costs have short-term impact, while indirect costs have long-term impact

4. What is the purpose of impact analysis approach?

- A) To identify the root cause of missing quality
- B) To determine the probability of missing quality events
- C) To identify the consequences of missing quality
- D) To calculate the financial cost of missing quality

5. What is the equation for calculating risk exposure in risk analysis approach?

- A) Risk exposure = Risk probability ÷ Potential loss
- B) Risk exposure = Risk probability + Potential loss
- C) Risk exposure = Risk probability × Potential loss
- D) Risk exposure = Risk probability - Potential loss

6. What is the IEEE level of risk for a situation with continuous usage, reversible environmental damages, and significant economic impact?

- A) Level B
- B) Level C
- C) Level A
- D) Level D

7. What is the purpose of a quality management system?

- A) To complete software projects within budget
- B) To document toxic requirements
- C) To improve the user interface of software products
- D) To achieve desired standard of quality

8. According to the concept of People - Process - Technology, what are the three leverage points for implementing quality mindset?

- A) Technology, Talent, Time
- B) People, Process, Technology
- C) Talent, Time, Tools
- D) Process, Technology, Talent

9. What does a quality management system ensure?

- A) Documentation of processes, policies, and workflows
- B) Reduction of direct costs
- C) Effective risk analysis approach
- D) High software development budget

10. Why is conformance to toxic requirements not equivalent to quality?

- A) Conformance to toxic requirements is costly
- B) Quality cannot be measured objectively
- C) Toxic requirements are difficult to understand
- D) Toxic requirements are filled with defects

11. What does direct cost of missing quality mean?

- A) Financial loss that occurs immediately after an event
- B) Effects that are easily observable/measurable
- C) Visible and tangible consequences
- D) Physical injury caused by an unfortunate event

12. What is an example of indirect cost of missing quality?

- A) Financial loss due to a catastrophic event
- B) Physical injuries resulting from direct costs
- C) Visible and measurable consequences
- D) Loss of market share or reputation

13. What is the purpose of risk analysis approach in determining the cost of missing quality?

- A) To calculate the direct cost of missing quality
- B) To evaluate the probability and potential loss of missing quality events
- C) To identify the root cause of missing quality
- D) To improve the reputation of organizations

14. What is the impact of missing quality attributes in software solutions?

- A) Customers lose trust in software solutions
- B) Suppliers suffer financial loss
- C) Both customers and suppliers bear consequences
- D) Customers face cessation in business operation

15. What does the intensity or impact of the loss depend on?

- A) The level of criticality of the software solution
- B) The financial cost of damage control
- C) The probability of missing quality events
- D) The time and place of missing quality events

16. What is the potential loss in the risk analysis approach?

- A) The total cost of missing quality events
- B) The risk probability in the analyzed situation
- C) The root cause of missing quality
- D) The impact or potential loss of a risk

17. What does risk exposure represent in the risk analysis approach equation?

- A) The product of the risk probability and potential loss
- B) The division of the risk probability and potential loss
- C) The sum of the risk probability and potential loss
- D) The subtraction of the risk probability and potential loss

18. What is the importance of impact analysis approach?

- A) To measure the intensity of the loss
- B) To perform root-cause analysis and identify problem
- C) To calculate the financial impact of missing quality
- D) To improve the user interface of software products

19. What is the potential impact of missing quality in a software product?

- A) Increased market share and reputation
- B) More threats and insecurity
- C) Enhanced user interface and usability
- D) More budget and better quality

20. What is the impact of lack of quality in software products?

- A) Maximize financial resources
- B) Promote cosmetic changes in UI
- C) Expect catastrophe
- D) Improve database security

21. What risk level signifies major financial loss, irreversible environmental damages, and continuous usage?

- A) Level C
- B) Level B
- C) Level A
- D) Level D

22. What classifies the costs of missing quality as direct and indirect?

- A) Eppler and Helfert principles
- B) CMU/SEI-93-TR-59
- C) ISO/IEC 9126-1
- D) IEEE Standard for Software Verification and Validation

23. What happened to the sale price of Yahoo when the breach announcement was made?

- A) Dropped by \$350 million
- B) Dropped by \$4.8 billion
- C) Remained the same
- D) Increased by \$350 million

24. What did Verizon negotiate to purchase Yahoo for?

- A) \$4.8 billion
- B) \$4.48 billion
- C) \$350 million
- D) \$350 billion

25. When did the breach incident with Yahoo occur?

- A) Late 2014
- B) Late 2016
- C) 2020
- D) Early 2014

26. What did Nokia's CEO mention in his farewell speech?

- A) We forgot to prioritize customer satisfaction
- B) We didn't do anything wrong but somehow we Lost
- C) We failed to invest in quality management system
- D) We couldn't meet toxic requirements

Answer Sheet

1. C
2. A
3. C
4. C
5. C
6. A
7. D
8. B
9. A
10. D
11. B
12. D
13. B
14. C
15. A
16. D
17. A
18. B
19. B
20. C
21. C
22. A
23. A
24. A
25. A
26. B

1. What is the function of the Process in the People-Process-Technology triad?

- A) To ensure synergy among people and technology
- B) To provide high-leverage focus on quality
- C) To realize immediate results through its magic
- D) To act as a glue and keep together the other two aspects

2. What is the purpose of a process?

- A) To create a magic stick for immediate results
- B) To achieve a given objective in a uniform and standardized manner
- C) To provide synergy between people and technology
- D) To ensure optimal utilization of resources

3. Why is a well-defined and implemented process important?

- A) It ensures stability in results and maximizes hard work
- B) It relies solely on the skills and training of the workforce
- C) It guarantees immediate and magical results
- D) It enables working smarter and utilizing people and technology at an optimal level

4. What is process benchmarking?

- A) Implementing external processes without evaluating internal limitations and resources
- B) Documenting critical elements of a process such as responsibility definition and process flow
- C) Comparing existing processes and performance metrics to industry's best practices
- D) Determining good or bad processes based on internal circumstances

5. What is the main purpose of documenting processes?

- A) To achieve stability and proactivity in work
- B) To improvise processes during projects
- C) To avoid confusion and provide clear definitions of responsibilities
- D) To sacrifice quality for unrealistic budget and schedule

6. What distinguishes mature organizations from immature organizations?

- A) Mature organizations have no objective measures of quality
- B) Mature organizations rely on documented processes and have well-defined roles and responsibilities
- C) Mature organizations focus on gut feelings rather than documented processes
- D) Imperfect processes are solely found in mature organizations

7. What is Capability Maturity Model Integration (CMMI)?

- A) A model that emphasizes massive changes to bring high quality
- B) A process that directs the work of software development
- C) A collection/model of best practices in systems, product, and software development
- D) A model that defines the roadmap for software development

8. What is the concept of maturity levels in CMMI?

- A) Quantitative understanding and control of software processes
- B) Evaluation criteria for measuring behavioral effectiveness
- C) Successive foundations for continuous process improvement
- D) Sequential stages of system, product, and software development

9. What belief does CMMI hold in terms of change and improvement?

- A) Change should primarily focus on crisis recovery
- B) Change should only occur in crisis situations
- C) Change should be normal, incremental, and future-oriented
- D) Change should be massive and immediate

10. What is the behavior of an organization at CMMI Maturity Level 1?

- A) Proactive and future-oriented with frequent change management
- B) Predictable and structured with a roadmap for software development
- C) Unstable and unpredictable with imperfectly defined and ad hoc processes
- D) Stable and controlled with perfect adherence to defined processes

11. In CMMI Maturity Level 1, what issues may organizations face?

- A) Difficulties in retaining talented resources and uncertainty in the organization
- B) Resistance to change and incremental improvements
- C) Inability to incorporate software engineering practices
- D) Lack of benefits from effective planning

12. In the scenario mentioned, what is the state of the software development process?

- A) Ad hoc with no roadmap and uncertainty
- B) Structured with a well-defined project plan
- C) Optimal and driven by client satisfaction
- D) Completely reactive and changeable

Answer Sheet

1. D
2. B
3. D
4. C
5. C
6. B
7. C
8. C
9. C
10. C
11. A
12. A

1. At which CMMI maturity level are the policies and frameworks established for software development projects?

- A) Maturity Level 3 - Defined
- B) Maturity Level 4 – Quantitatively Managed
- C) Maturity Level 5 – Optimized
- D) Maturity Level 2 – Managed

2. What is the major difference between CMMI Maturity Level 2 and Level 3?

- A) The process, standards, and procedures for a project are tailored from the organization's set of standard processes to suit a particular project or organizational unit.
- B) Level 2 is highly predictable while Level 3 is optimized for continuous improvement.
- C) Level 2 focuses on reactive processes while Level 3 focuses on defined processes.
- D) Level 2 relies heavily on Heroes while Level 3 relies on defined processes.

3. What happens at CMMI Maturity Level 3?

- A) Software process, quality, and productivity metrics are measured and analyzed.
- B) Continuous process improvement is the focus of the organization.
- C) Quantitative objectives are established to evaluate process performance.
- D) The standard processes for developing and maintaining software are established and documented.

4. At CMMI Maturity Level 4, what can management measure using quantitative techniques?

- A) Special variation points for further improvements.
- B) The predictability of process performance.
- C) Different valuable metrics like software process, quality, and productivity.
- D) The quality and process performance statistically.

5. What is the focus at CMMI Maturity Level 5?

- A) Continuous process improvement and striving for incremental changes in the process capability and performance.
- B) Controlling the performance of processes using statistical and quantitative techniques.
- C) Achieving control over products and processes by narrowing process performance variation.
- D) Addressing special causes of variation and providing statistical predictability of results.

6. How many Capability Levels are there in CMMI?

- A) Three
- B) Five
- C) Seven
- D) Six

7. What do the Capability Levels represent in CMMI?

- A) The maturity level of the organization.
- B) The effectiveness of the organization's processes.
- C) The different stages of process improvement.
- D) How well an organization is aligned to a specific process area.

8. Which component of the CMMI process model includes a cluster of related practices in an area?

- A) Capability Level
- B) Generic Practices
- C) **Process Area**
- D) Specific Practice

9. What are Generic Practices in the CMMI process model?

- A) **Expected model components that contribute to the institutionalization** of the processes associated with a process area.
- B) Expected model components that contribute to process improvement.
- C) Expected model components related to achieving specific goals.
- D) Clusters of related practices in an area.

10. What are Specific Practices in the CMMI process model?

- A) Expected model components related to achieving generic goals.
- B) Expected model components that contribute to the institutionalization of the processes associated with a process area.
- C) **Expected model components that describe the activities expected to** result in the achievement of specific goals of a process area.
- D) Clusters of related practices in an area.

11. How many process areas are there in CMMI?

- A) 24
- B) 30
- C) 18
- D) 12

12. What is the optimal level of each process area achieved in?

- A) Increments
- B) Continuous
- C) Stages
- D) Standard

13. In the continuous representation of process areas, what does an organization do?

- A) Adopts new technologies and methods to improve process capability.
- B) Focuses on incremental changes in the process capability and performance.
- C) Achieves the goals of the process areas associated with each maturity level.
- D) Chooses the processes critical to its business and achieves high capability levels.

14. In the staged representation of process areas, what does an organization do?

- A) Focuses on incremental changes in the process capability and performance.
- B) Chooses the processes critical to its business and achieves high capability levels.
- C) Achieves the goals of the process areas associated with each maturity level.
- D) Adopts new technologies and methods to improve process capability.

15. At which level of CMMI is significant focus given to addressing common causes of variation and changing the process to improve performance?

- A) Level 3
- B) Level 2
- C) Level 4
- D) Level 5

16. What is the significance of statistical analysis at CMMI Maturity Level 4?

- A) Predictions are based on a statistical analysis of fine-grained process data.
- B) Software process, quality, and productivity metrics are measured and analyzed.
- C) Quantitative boundaries are decided for processes to achieve control and improve variation.
- D) Quantitative objectives are used to evaluate the quality and process performance.

17. Why do organizations at CMMI Maturity Level 5 earn the ability to proactively evaluate processes?

- A) To address special causes of variation and provide statistical predictability.
- B) To avoid defects and strive for continuous process improvement.
- C) To manage processes quantitatively and narrow the process performance variation.
- D) To establish and document standard processes for development and maintenance.

18. What is the characteristic of activities at CMMI Maturity Level 2?

- A) Processes are managed with established policies and frameworks.
- B) Processes are standardized and consistent, with clearly defined roles and responsibilities.
- C) Processes are often reactive, relying heavily on Heroes.
- D) Processes are quantitatively managed to evaluate quality and process performance.

19. What is the characteristic of activities at CMMI Maturity Level 3?

- A) Processes are well-defined and documented, with standard plans and work procedures.
- B) Processes are managed with established policies and frameworks.
- C) Processes are quantitatively managed to evaluate quality and process performance.
- D) Processes are optimized for continuous improvement and incremental changes.

20. What is the characteristic of activities at CMMI Maturity Level 4?

- A) Processes are well-defined and documented, with standard plans and work procedures.
- B) Processes are standardized and consistent, with clearly defined roles and responsibilities.
- C) Processes are optimized for continuous improvement and incremental changes.
- D) Processes are statistically managed and fine-grained data is used to control process performance.

21. What is the characteristic of activities at CMMI Maturity Level 5?

- A) Processes are statistically managed and fine-grained data is used to control process performance.
- B) Processes are optimized for continuous improvement by addressing common causes of variation.
- C) Processes are quantitatively managed to evaluate quality and process performance.
- D) Processes are standardized and consistent, with clearly defined roles and responsibilities.

22. In the given example for CMMI Maturity Level 2, what was the blocker issue faced by the Software House during a meeting?

- A) One of the Software Architect was absent with no other resource available.
- B) The company's credibility level went down to zero.
- C) There was no documentation except the Software Architect.
- D) The client from Indonesia stopped using the Product.

23. In the given example for CMMI Maturity Level 3, what is the to-do list for the Project Manager after project assignment?

- A) Attending a meeting with the software house's marketing team and development team.
- B) Ensuring the availability of adequate resources and training for the workforce.
- C) Discussing requirements of a new module requested by the Indonesian client.
- D) Internal Kickoff to discuss and clarify scope-related queries, Client Kickoff including team introduction and approval of scope by the client.

24. In the given example for CMMI Maturity Level 4, what is the lesson learned?

- A) Process without statistics cannot be improved.
- B) Process should focus on addressing special causes of variation.
- C) Process improvement requires the narrowing of process performance variation.
- D) Process should be predictable and controllable using statistical techniques.

25. In the given example for CMMI Maturity Level 5, what is one option for further optimization?

- A) Identifying special variation points for further improvements.
- B) Adopting new technologies and methods for continuous process improvement.
- C) Providing statistical predictability of process performance.
- D) Reducing the audience of the meeting

26. What is the difference between CMMI Maturity Level 4 and Level 5?

A) Level 4 provides statistical predictability of results, while Level 5 maintains statistical predictability of process performance.

B) Level 4 focuses on addressing special causes of variation, while Level 5 addresses common causes of variation and changing the process to improve performance.

C) Level 4 uses statistical techniques for controlling process performance, while Level 5 focuses on continuous process improvement.

D) Level 4 manages processes quantitatively to achieve control, while Level 5 strives for incremental changes in process capability.

Answer Sheet

1. D
2. A
3. D
4. C
5. A
6. D
7. D
8. C
9. A
10. C
11. A
12. A
13. D
14. C
15. D
16. A
17. B
18. C
19. A
20. D
21. B
22. A
23. D
24. A
25. D
26. B

1. What is the major purpose of the Engineering Process Group (EPG)?
 - A) To define what a process should be
 - B) To evaluate the existing process
 - C) To improve the process throughout the organization
 - D) To manage multiple process action teams

2. What are Process Action Teams (PAT) responsible for?
 - A) Evaluating the existing process
 - B) Defining what a process should be
 - C) Implementation of improvement initiatives activities in specific process areas
 - D) Managing the entire organization's process

3. What do PATs focus on during the evaluation stage?
 - A) Strengths found in the process
 - B) Technical staff from throughout the organization
 - C) Existing process documentation
 - D) Weaknesses found in the process

4. Which process areas will be discussed in the course?

- A) All 22 process areas
- B) Engineering related process areas
- C) Non-engineering related process areas
- D) Technical solution process areas

5. What is the role of the Process Action Teams (PATs)?

- A) To develop project plans and work products
- B) To write, pilot, and update process improvement procedures
- C) To evaluate the existing process
- D) To manage multiple process areas

6. What is the task of the Process Action Teams (PATs)?

- A) Creating project plans and work products
- B) Generating process improvement documentation
- C) Managing different stakeholders
- D) Implementing improvement initiatives

7. What is the goal of Requirement Management process area?

- A) To ensure alignment between requirements, project plans, and the final output
- B) To maintain bidirectional traceability of requirements
- C) To manage different versions of requirements
- D) To manage inconsistencies between products and requirements

8. What is the purpose of Requirement Development process area?

- A) To develop software requirement specifications
- B) To manage inconsistencies between products and requirements
- C) To ensure alignment between requirements, project plans, and the final output
- D) To analyze and establish customer, product, and product component requirements

9. What does Requirement Development process area involve?

- A) Identifying and refining requirements throughout the product lifecycle
- B) Collecting and translating stakeholder needs into requirements
- C) Elaborating customer requirements into product requirements
- D) Documenting and analyzing requirements

10. What is the primary artifact for Requirement Development process area?

- A) Standardized templates
- B) Procedures and policies
- C) Development of Software Requirement Specification (SRS)
- D) Action Plans

11. What needs to happen for an SRS to be considered acceptable?

- A) It must be implemented by the project managers
- B) It must be explicitly approved by the client
- C) It must be drafted by the Process Action Teams (PATs)
- D) It must include standardized templates

12. What is one of the motivations behind Requirement Management process area?

- A) To ensure traceability between source and lower level requirements
- B) To manage correlation between project deliverables and requirements
- C) To manage inconsistencies between products and requirements
- D) To manage different versions of requirements

13. What is the purpose of managing requirements changes in Requirement Management process area?

- A) To obtain commitment to requirements from project stakeholders
- B) To manage inconsistencies between products and requirements
- C) To maintain bidirectional traceability of requirements
- D) To manage changes to requirements as they evolve during the project

14. What is one of the practices involved in Requirement Development?

- A) Analyzing and validating requirements
- B) Developing product requirements
- C) Maintaining bidirectional traceability of requirements
- D) Developing customer requirements

15. What is one of the practices involved in Requirement Management?

- A) Understanding requirements
- B) Managing requirements changes
- C) Maintaining bidirectional traceability of requirements
- D) Obtaining commitment to requirements

16. What role does a traceability matrix play in Requirement Management?

- A) It ensures consistency in documentation throughout the organization
- B) It provides a structure for documenting requirements
- C) It is used to manage cross-referencing of requirements
- D) It traces the changes made to requirements

17. What is one of the responsibilities of the Process Action Teams (PATs)?

- A) Defining what a process should be
- B) Evaluating the existing process
- C) Bringing consistency in the documents throughout the organization
- D) Managing multiple process action teams

18. Who are the members of the Process Action Teams (PATs)?

- A) EPG members
- B) Technical staff from throughout the organizations
- C) Transition partners
- D) Project managers

19. How many process areas will be discussed in the course?

- A) 10
- B) 22
- C) 6
- D) 4

20. What are PATs also known as?

- A) Technical staff
- B) Process improvement experts
- C) Worker bees
- D) Process action teams

21. What is the task of Process Action Teams (PATs) in improvement initiatives?

- A) Take care of different stakeholders for different process areas
- B) Draft standardized templates for processes
- C) Generate process improvement documentation
- D) Bring consistency in the documents throughout the organization

22. Why do Process Action Teams (PATs) need to work on drafting templates?

- A) To bring consistency in the documents throughout the organization
- B) To manage different stakeholders
- C) To avoid rewriting of documents
- D) To generate process improvement procedures

23. What is a key task of Process Action Teams (PATs)?

- A) Management of multiple process action teams
- B) Generation of process improvement documentation
- C) Standardization of artifacts
- D) Identification of weaknesses in the process

24. What do Technical Solution process area and Product Integration process area cover?

- A) Software validation and verification activities
- B) Implementation and deployment activities
- C) Development and maintenance activities shared across engineering disciplines
- D) Requirement management and analysis activities

25. What are the two types of changes in the requirement?

- A) Completion of partial requirement and changes in requirement priority
- B) Update in existing requirement and addition of new requirement
- C) Consolidation of multiple requirements and division of complicated requirement
- D) Update in existing requirement and removal of unnecessary requirement

26. In the example of Requirement Management, what caused the rejection of the Demo?

- A) Inconsistencies between products and requirements
- B) Different versions of requirements
- C) Lack of commitment to requirements from project stakeholders
- D) Inadequate management of requirements changes

Answer Sheet

1. C
2. C
3. D
4. B
5. B
6. B
7. A
8. D
9. A
10. C
11. B
12. C
13. D
14. A
15. C
16. C
17. C
18. B
19. C
20. C
21. A
22. A
23. C
24. C
25. B
26. B

1. What is the purpose of the Technical Solutions process area?
 - A) Evaluation of different solution or design approaches.
 - B) Streamlining the requirements, project plans, and final output.
 - C) Visualizing the technical design using wireframes.
 - D) Selection, design, and implementation of solutions to the requirement of the product/project.

2. Which of the following is a goal of the Technical Solutions process area?
 - A) Select Product Component Solutions: Product or product component solutions are selected from alternative solutions.
 - B) Develop the Design: Product or product component designs are developed.
 - C) Implement the Product Design: Product components, and associated support documentation, are implemented from their designs.
 - D) Documenting the technical design in the Technical Design Document.

3. What is a major risk in the Product Integration process area?
 - A) Product components not integrating with each other.
 - B) Integrating all the components simultaneously.
 - C) Heterogeneous development environment.
 - D) Quality requirements not being satisfied.

4. What is the purpose of the Software Validation process area?

- A) Assembling and delivering the product components.
- B) Ensuring that the final product or its component(s) fulfill the requirements and its intended use when deployed.
- C) Selection and design of solutions to the requirement of the product/project.
- D) Preparing for validation by selecting the product, validation environment, and criteria.

5. What is a major goal of the Software Validation process area?

- A) Prepare for Validation: Preparation for validation is conducted by selecting the product, validation environment, and validation criteria.
- B) Building the right thing based on client requirements.
- C) Validate Product or Product Components: The product or product components are validated to ensure they are suitable for use in their intended operating environment.
- D) Verifying product components are assembled and delivering the integrated product.

6. Which type of testing is performed to ensure successful completion of integration?

- A) Functional testing
- B) **Sanity testing**
- C) Regression testing
- D) Performance testing

7. What is the main artifact in the Technical Solutions process area?

- A) Appendix - I
- B) WireFrames
- C) Product Integration
- D) **Technical Design Document**

8. What is the purpose of the Technical Design Document?

- A) To assemble product components and deliver the product.
- B) To **streamline the requirements, project plans**, and final output (product).
- C) To evaluate different solution or design approaches.
- D) To capture client requirements and build the right thing.

9. What does the Technical Design Document provide a picture of?

- A) Adopted technical design
- B) Traceability with the requirement
- C) Product architecture
- D) Product components

10. What major failure can occur in the Product Integration process area?

- A) Defects due to misaligned interfaces.
- B) Assembling product components incrementally.
- C) Heterogeneous development environment.
- D) Product components failed to integrate with each other or partially integrate.

11. What are the main goals for the Technical Solutions process area?

- A) Evaluate different solution or design approaches, Visualize the technical design using wireframes
- B) Prepare for Product Integration, Ensure Interface Compatibility, Assemble Product Components and Deliver the Product
- C) Select Product Component Solutions, Develop the Design, Implement the Product Design
- D) Prepare for Validation, Validate Product or Product Components

12. How is sanity testing typically performed in product integration?

- A) To select product, validation environment, and validation criteria.
- B) To remove issues on the merged codebase.
- C) To ensure integration is successfully completed and no issues/defects are introduced due to it.
- D) To verify the compatibility of product component interfaces.

13. What is the purpose of preparing for validation in the Software Validation process area?

- A) Building the right thing based on client requirements.
- B) Ensuring final product or its component(s) fulfill the requirements and its intended use.
- C) Validating the product or product components in the intended operating environment.
- D) Selecting the product, validation environment, and validation criteria.

14. What is the major goal of the Technical Solutions process area?

- A) Evaluation of different solution or design approaches.
- B) Implement the Product Design: Product components, and associated support documentation, are implemented from their designs.
- C) Select Product Component Solutions: Product or product component solutions are selected from alternative solutions.
- D) Develop the Design: Product or product component designs are developed.

15. What components are integrated and tested first in product integration?

- A) All the components simultaneously
- B) Only few components
- C) Product components not integrating with each other
- D) Defects due to misaligned interfaces

16. Which of the following is considered a key aspect of software validation?

- A) Proof of Concept
- B) Software Configuration
- C) Component Integration
- D) Technical Design Document

17. What are the main goals for the Software Validation process area?

- A) Select Product Component Solutions, Develop the Design, Implement the Product Design
- B) Prepare for Validation, Validate Product or Product Components
- C) Prepare for Product Integration, Ensure Interface Compatibility, Assemble Product Components and Deliver the Product
- D) Evaluation of different solution or design approaches, Visualize the technical design using wireframes

18. What is the intended use of software validation?

- A) To evaluate different solution or design approaches.
- B) To select and design solutions to the requirement of the product/project.
- C) To ensure the product is doing what it should be doing (as per requirements) in the intended environment.
- D) To assemble product components and deliver the product.

19. What is the purpose of the Product Integration process area?

- A) Validating the product or product components in the intended operating environment.
- B) Selection, design, and implementation of solutions to the requirement of the product/project.
- C) Preparation for validation by selecting the product, validation environment, and criteria.
- D) Assembling the product from multiple product components and ensuring that the product behaves properly and satisfies all the functional and quality requirements.

20. What is a major goal of the Technical Solutions process area?

- A) Develop the Design: Product or product component designs are developed.
- B) Visualizing the technical design using wireframes.
- C) Select Product Component Solutions: Product or product component solutions are selected from alternative solutions.
- D) Implement the Product Design: Product components, and associated support documentation, are implemented from their designs.

21. What is the main artifact in the Product Integration process area?

- A) Sanity testing
- B) Merged codebase
- C) Product Design Document
- D) Software Configuration

22. What does the Sample Technical Design Document capture?

- A) Evaluation and comparison of different solution or design approaches.
- B) Preparation for product integration, ensuring internal and external interface compatibility.
- C) Wireframes to visualize the technical design.
- D) Details of adopted technical design, product architecture, and traceability with the requirement.

23. What is a goal of the Product Integration process area?

- A) Assemble Product Components and Deliver the Product: Verified product components are assembled and the integrated, verified, and validated product is delivered.
- B) Remove issues on the merged codebase.
- C) **Ensure Interface Compatibility: The** product component interfaces, both internal and external, are compatible.
- D) Prepare for Product Integration: Preparation for product integration is conducted.

24. What is the main purpose of the Technical Solutions process area?

- A) Validating the product or its components in the intended operational environment.
- B) Selecting, designing, and implementing solutions to the requirement of the product/project.
- C) **Evaluating different solution or design** approaches that satisfy functional and quality requirements.
- D) Ensuring that the product components are integrated properly.

25. What is the primary purpose of the Technical Design Document?

- A) To assemble and deliver product components from their designs.
- B) To visualize the technical design using wireframes.
- C) To document the details and traceability of the adopted technical design.
- D) To streamline the requirements, project plans, and final output (product).

Answer Sheet

1. D
2. B
3. C
4. B
5. C
6. B
7. D
8. B
9. C
10. D
11. C
12. C
13. D
14. A
15. B
16. A
17. B
18. C
19. D
20. B
21. B
22. D
23. C
24. C
25. C

1. What is the purpose of the Software Verification process area?

- A) To build the product in the right way
- B) To ensure the final product is error-free and meets specified requirements
- C) To evaluate the usefulness of the system
- D) To assess compliance with quality standards

2. Which activities are included in software verification?

- A) Testing, design analysis, inspections, and code reviews
- B) Customer approval, prototype development, and product delivery
- C) Assessment and process improvement
- D) Software registration, attendance marking, and login

3. What are prototypes used for in the software validation process?

- A) To ensure that the product is error-free
- B) To evaluate the usefulness of the system
- C) To show to the customer and get approval for the actual product
- D) To build the product in the right way

4. Which goals are associated with the Software Verification process area?

- A) Coordinating process activity throughout the organization
- B) Prepare for Verification: Preparation for verification is conducted.
Verify Selected Work Products: Selected work products are verified against their specified requirements.
- C) Implementing software processes to ensure compliance
- D) Measuring performance or compliance level

5. What is the purpose of the Engineering Process Group?

- A) To assess existing processes and provide suggestions for improvement
- B) To measure the organizational compliance level with established processes
- C) To implement software processes, monitor, evaluate, and improve processes, and coordinate process activity throughout the organization
- D) To conduct audits and evaluate conformance

6. What is the role of audits in organizational processes?

- A) To evaluate the usefulness of the system
- B) To assess compliance with quality standards
- C) To ensure the final product is error-free
- D) To measure compliance level, identify reasons for nonconformance, and provide insights for process improvement

7. Who conducts audits?

- A) The engineering process group
- B) The QA team
- C) Independent auditors
- D) The customer or contracted organization

8. What is the purpose of audits?

- A) To assess compliance with quality standards
- B) To evaluate the usefulness of the system
- C) To check compliance, evaluate reasons for nonconformance, and provide a starting point for process improvement
- D) To ensure the final product is error-free

9. What are the three types of audits?

- A) First Party Audits, Second Party Audits, Third Party Audits
- B) Internal Audits, External Audits, Supplier Audits
- C) Administrative Audits, Technical Audits, QA Audits
- D) Developer Audits, Tester Audits, Manager Audits

10. Who leads the audit activities in an audit team?

- A) Developer Lead
- B) Certified lead auditor
- C) QA Lead
- D) Project Manager

11. What is the purpose of CMMI appraisals?

- A) To ensure the final product is error-free
- B) To assess compliance with quality standards
- C) To prepare for verification
- D) To collect, review, and analyze data to measure performance or compliance level

12. What is the frequency of process reviews in organizations?

- A) Biannual, quarterly, or on a need basis
- B) Annually
- C) Weekly
- D) Monthly

13. Why are process reviews important in organizations?

- A) To measure the effectiveness of the process and identify required actions for improvement
- B) To assess compliance with quality standards
- C) To evaluate the usefulness of the system
- D) To build the product in the right way

14. What is the purpose of a review policy?

- A) To evaluate the effectiveness of processes
- B) To define what and when to review
- C) To identify reasons for nonconformance
- D) To measure compliance level

15. What is benchmarking in the context of process review?

- A) Evaluating the usefulness of the system
- B) Referencing successful solutions from other organizations facing similar issues
- C) Ensuring compliance with quality standards
- D) Adopting best practices without modifications

16. What is the role of the audit team members?

- A) To ensure the final product is error-free
- B) To implement software processes
- C) To assess compliance with quality standards
- D) To measure process results and perform process audits for compliance

17. What does an audit measure?

- A) The usefulness of the system
- B) The quality of selected work products
- C) Performance or compliance level
- D) The effectiveness of the audit process

18. What are audits used for in a volatile working environment?

- A) To measure the compliance level with quality standards
- B) To maintain check and balance on organizational process and practices
- C) To assess the performance of employees
- D) To build the product in the right way

19. What type of audit is an internal audit?

- A) Supplier Audit
- B) First Party Audit
- C) Third Party Audit
- D) Second Party Audit

20. What type of audit is performed by a customer on a supplier?

- A) Second Party Audit
- B) Third Party Audit
- C) First Party Audit
- D) Internal Audit

21. What type of audit is performed by an audit organization independent of supplier-customer relationship?

- A) Third Party Audit
- B) First Party Audit
- C) External Audit
- D) Second Party Audit

22. What components are involved in the software verification process?

- A) Employee login and registration
- B) Selected work products and their specified requirements
- C) System usefulness and engineering aspects
- D) QA team and development team

23. What are the major goals of the Software Verification process area?

- A) Building the product and delivering it
- B) Prepare for Verification and Verify Selected Work Products
- C) Coordinating process activity and measuring compliance level
- D) Testing the product and analyzing design

24. What is the purpose of inspections and code reviews in software verification?

- A) To assess compliance with quality standards
- B) To ensure the final product is error-free
- C) To verify selected work products against their specified requirements
- D) To evaluate the usefulness of the system

25. What is the purpose of process appraisals?

- A) To assess compliance with quality standards
- B) To ensure the final product is error-free
- C) To prepare for verification
- D) To measure the capability or maturity level of the processes in use

26. What is the impact of process reviews on organizational processes?

- A) Evaluating the usefulness of the system
- B) Ensuring compliance with quality standards
- C) Identifying required actions for process improvement
- D) Coordinating process activity throughout the organization

Answer Sheet

1. B
2. A
3. C
4. B
5. C
6. D
7. C
8. C
9. A
10. B
11. D
12. A
13. A
14. B
15. B
16. D
17. C
18. B
19. B
20. A
21. A
22. B
23. B
24. C
25. D
26. C

